INCH-POUND

MIL-C-24643/44C
14 March 1994
SUPERSEDING
MIL-C-24643/44B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLT, TYPE LS3SP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

	First	- Copper conductor, AWG no. 18, class B stranding (7 by 0.0152 inch), uncoated. Maximum diameter 0.050 inch.
	Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
*	Third	- Crosslinked polyethylene insulation. Nominal wall thickness 0.020 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white and one red conductor for each triad.
	Fourth	- Three conductors (one black, one white and one red) cabled together with a nominal lay of 1.5 inches to form a triad.
*	Fifth	- One polyester tape, type G in accordance with MIL-I-631.
*	Sixth	- Braided shield of uncoated or tin coated copper. Minimum coverage of 85 percent.
*	Seventh	- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, an alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

AMSC N/A FSC 6145

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Construction (non-watertight) (continued)

* Eighth - The specified number of triads (see table I) cabled together with a lay not greater than 16 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. A single triad shall not be used as the central core of the cable. Fillers shall be used as necessary to form a firm, well rounded assembly.

Ninth - Binder tape, applied helically with overlap.

* Tenth - Cross-linked polyolefin jacket double layer, reinforced. Minimum average wall thickness 0.080.

TABLE I. Details.

Military part no. M24643/44	Type and size	Number of triads	Overall diameter (inches)	
			min	max
-01UN	LS3SF-7	7	1.000	1.040

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum ..	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor, within triads	2000
Shield to shield, between triads	500
Insulation resistance - Megohms/1000 feet minimum	
Conductor to conductor	500
* Shield to shield	100
Conductor to shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum	30
* Capacitance unbalance - percent minimum	10
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - at minus 30 ± 2°C, mandrel diameter 12 times cable diameter	No damage
Bending endurance - at 25± 2°C, cycles, minimum	25
Tear - pounds per inch thickness, minimum (ASTM D 470) ..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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EXAMINATION AND TESTS (continued):	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
* Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, minimum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet (nominal)

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(Project No. 6145-2057-44)

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